

EMPLOYEE MANAGEMENT SYSTEM PROJECT REPORT

Project Title:

Employee Management System Using Python

Introduction:

The Employee Management System is a console-based Python application developed to manage employee information efficiently. The program allows users to register employees, log in securely, view employee details, update employee information, change passwords, and delete employee records. This project demonstrates how Python can be used to build a simple real-world employee management application.

Objective:

The objectives of this project are:

- To develop a simple employee management system.
- To store employee information efficiently.
- To perform employee registration and login.
- To update employee details easily.
- To improve Python programming skills through a real-world project.

Software Used:

- Programming Language: Python
- IDE: Visual Studio Code
- Operating System: Windows

Python Concepts Use:

- Variables
- Data Types
- Dictionaries
- Functions
- Loops (while)
- Conditional Statements (if, elif, else)
- User Input
- Return Statements
- Nested Loops
- CRUD Operations (Create, Read, Update, Delete)

Project Working:

- The application starts by displaying the main menu.
- The user can register a new employee.
- A unique Employee ID is generated automatically.
- The employee logs in using Employee ID and Password.
- After successful login, the employee menu is displayed.
- The employee can view profile details.
- The employee can update department information.
- The employee can update designation.
- The employee can update salary details.
- The employee can change the login password.
- The employee can delete the account if required.
- The employee logs out safely and returns to the main menu.

Features:

- Employee Registration
- Employee Login Authentication
- View Employee Profile
- Update Department
- Update Designation
- Update Salary
- Change Password
- Delete Employee Record
- Logout Facility
- Console-Based Menu-Driven Interface

What I Learned:

Through this project, I learned how to use Python dictionaries to store employee information, functions to organize the program, loops for menu-driven execution, conditional statements for decision-making, and CRUD operations to manage employee records. I also improved my understanding of authentication and modular programming.

Future Improvements:

- Add Employee Attendance Management.
- Add Leave Management System.
- Store employee records in a MySQL database.
- Generate Employee ID cards.
- Export employee details to Excel or PDF.
- Add Admin and Employee roles.
- Develop a graphical user interface (GUI) using Tkinter.
- Integrate the project with a web application using Django or Flask.

Conclusion:

The Employee Management System is a simple and efficient Python project that automates employee record management. It minimizes manual work, improves data organization, and provides practical experience with Python programming concepts. This project strengthened my understanding of menu-driven applications and real-world software development using Python.

REGISTER

```
def register(name, email, department, designation, salary, password):
```

```
    emp_id = max(employees.keys()) + 1
```

```
    employees[emp_id] =  
        { "name": name,  
          "email": email,  
          "department": department,  
          "designation": designation,  
          "salary": salary,  
          "password": password  
        }
```

```
    print("\nEmployee Registered Successfully")  
    print("Employee ID :", emp_id)
```

LOGIN

```
def login(emp_id, password):
```

```
if emp_id in employees:

    if employees[emp_id]["password"] == password:
        print("\nLogin Successful")
        return True

    else:
        print("Incorrect Password")
        return False

else:
    print("Employee Not Found")
    return False
```

VIEW PROFILE

```
def view_profile(emp_id):

    emp = employees[emp_id]

    print("\n===== EMPLOYEE PROFILE =====")
    print("Employee ID :", emp_id)
    print("Name      :", emp["name"])
    print("Email      :", emp["email"])
    print("Department :", emp["department"])
    print("Designation :", emp["designation"])
    print("Salary      :", emp["salary"])
    print("=====")
```

UPDATE DEPARTMENT

```
def update_department(emp_id, department):

    employees[emp_id]["department"] = department

    print("Department Updated Successfully")
```

UPDATE DESIGNATION

```
def update_designation(emp_id, designation):  
  
    employees[emp_id]["designation"] = designation  
  
    print("Designation Updated Successfully")
```

UPDATE SALARY

```
def update_salary(emp_id, salary):  
  
    employees[emp_id]["salary"] = salary  
  
    print("Salary Updated Successfully")
```

CHANGE PASSWORD

```
def change_password(emp_id, password):  
  
    employees[emp_id]["password"] = password  
  
    print("Password Changed Successfully")
```

DELETE EMPLOYEE

```
def delete_employee(emp_id):  
  
    del employees[emp_id]  
  
    print("Employee Deleted Successfully")  
  
    return False
```

LOGOUT

```
def logout():  
  
    print("Logout Successful")  
  
    return False
```

MAIN PROGRAM

```
if name == "main":  
  
    while True:  
  
        print("\n===== EMPLOYEE MANAGEMENT SYSTEM =====")  
        print("1. Register Employee")  
        print("2. Login")  
        print("3. Exit")  
  
        choice = int(input("Enter Your Choice : "))  
  
        if choice == 1:  
  
            name = input("Enter Name : ")  
            email = input("Enter Email : ")  
            department = input("Enter Department : ")  
            designation = input("Enter Designation : ")  
            salary = int(input("Enter Salary : "))  
            password = input("Create Password : ")  
  
            register(  
                name, email,  
                department,  
                designation,  
                salary,  
                password  
            )
```

elif choice == 2:

emp_id = int(input("Enter Employee ID : "))

password = input("Enter Password : ")

login_status = login(emp_id, password)

while login_status:

print("\n===== EMPLOYEE MENU =====")

print("1. View Profile")

print("2. Update Department")

print("3. Update Designation")

print("4. Update Salary")

print("5. Change Password")

print("6. Delete Employee")

print("7. Logout")

option = int(input("Enter Your Choice : "))

if option == 1:

view_profile(emp_id)

elif option == 2:

department = input("Enter New Department : ")

update_department(emp_id, department)

elif option == 3:

designation = input("Enter New Designation : ")

update_designation(emp_id, designation)

elif option == 4:


```
    salary = int(input("Enter New Salary : "))

    )

    update_salary(emp_id, salary)

elif option == 5:

    password = input("Enter New Password : ")

    change_password(emp_id, password)

elif option == 6:

    confirm = input("Are You Sure (yes/no) : ")

    if confirm.lower() == "yes":

        login_status = delete_employee(emp_id)

elif option == 7:

    login_status = logout()

else:

    print("Invalid Choice")

elif choice == 3:

    print("Thank You for Using Employee Management System")
    break

else:
```

```
print("Invalid Choice")
```